LESSON 4.9

FINDING THE AREA OF A CIRCLE

LESSON INTRODUCTION

MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

LEARNING TARGETS

- I can find the area of a circle given the radius or diameter.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

ESSENTIAL QUESTIONS

- How do you use the radius to find the area of a circle?
- How do you use the diameter to find the area of a circle?

LESSON NARRATIVE

In this lesson, students use the formula for the area of a circle to find that area of circles in real-world problems. Students will be exploring different approximations of pi and the slight difference in answers obtained based on the approximation utilized.

COMPONENT SUMMARY

4.9.1 WARM-UP *(~5 MIN.)*

Determine the most accurate drawing of a circle

4.9.2 GUIDED INSTRUCTION *(15 MIN.)*

Consolidate understanding of the formula for area of a circle

4.9.3 GUIDED PRACTICE *(10 MIN.)*

Explore impact of different approximations of pi on calculations of the areas of a circle

4.9.4 YOUR TURN! *(10 MIN.)*

Engage in calculating the areas of circles in real-world problems

4.9.5 WRAP-UP *(~5 MIN.)*

Identify and correct errors in samples of students' work

RESOURCES

GLOSSARY

Key Terms:

- Area of a circle

Additional Terms:

- N/A

MATERIALS

- (Optional) Chart paper & markers
- Calculator

ADDITIONAL PREPARATION

- Consider modifying real-world problems in 4.9.4 Your Turn! to include people, places, and things that connect to students' interests and everyday lives.

TEACHER TIPS

COMMON MISCONCEPTIONS

- In question 2 in 4.9.3 Guided Practice, students may struggle to determine the error if they forget that they need the radius instead of the diameter to solve for the area.

THINGS TO CONSIDER

- Consider having students create anchor charts or Frayer Models with the area formula, visuals, and examples.
- If time is an issue, then consider assigning Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 4.9.3 Guided Practice, prompt students to initially formulate their answers to the problems individually. Then, ask students to discuss their work and answers with a partner. Finally, debrief with a whole-class discussion in which a variety of students share their strategies.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 4.9.4 students apply the area of a circle formula to solve real-world problems involving a circular table and clock. Student interest will be ignited as they see the utility of the math they are learning. Students also engage with MA.K12.MTR.4.1 in the Wrap-Up: Error Analysis.

DIFFERENTIATE INSTRUCTION

In 4.9.4 Your Turn! consider providing an auditory entry point by reading the problems aloud or having students use the text reader in the online version of the lesson. Prompt students to underline key information they will need to solve the problems. Then prompt students to label the circle diagrams with the dimensions they know and variables for the unknown quantities. Prompt students to reference their Anchor Charts or Frayer Models with formulas, visuals, and examples.

4.9.4 Your Turn!

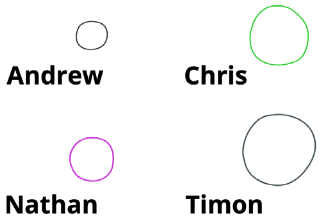
Elena wants to tile the top of a circular table. The radius of the top is $14\frac{5}{8}$ inches (in.). What is the area of the table that will be tiled? Use $\frac{22}{7}$ for π.

4.9.1 WARM-UP: WHICH IS THE BEST?

TEACHER MOVES

Students may struggle with determining how to rank the circles. Ask students to consider the definition of a circle and consider basing their ranking on how well the circles represent the definition.

4.9.1 Warm-Up: Which Is the Best?



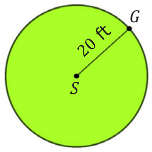
1. Rank each person's image based on which looks most like a circle.
   Answers will vary. Chris, Nathan, Timon, Andrew

2. Explain how you decided upon your ranking.
   Students should estimate the center of each figure and then determine if the points on the circle are equidistant from the center. Students may also have estimated the center and then drawn radii to see which circle has the most congruent radii.

4.9.2 GUIDED INSTRUCTION: FINDING THE AREA OF A CIRCLE

4.9.2 Guided Instruction: Finding the Area of a Circle

1. A goat, located at point G, is tied with a 20 foot (ft) rope to a stake located at point S.



   a. Complete each statement with words from the word bank.

$A = \pi r^2$	radius	$C = 2\pi r$	area	circumference

   The space in which the goat can roam is the <u>*area*</u> of the circle. The distance of the rope from the stake to the goat is the <u>*radius*</u> of the circle. To find the total space in which the goat can roam, use the formula $\underline{A = \pi r^2}$.

   b. What is the area over which the goat can roam?
   Answers may vary based on the approximation used for pi. See question 2.

The **area of a circle**, A, can be found using the formula $A = \pi r^2$ or $A = \frac{1}{2}Cr$, where C is the circumference of the circle, π is the proportional relationship between the diameter and the circumference of any circle, and r is the radius.

DIFFERENTIATE INSTRUCTION

Consider your observations from the previous lesson or the Wrap-Up to inform your approach to 4.9.2 Guided Instruction. Students explored the area of a circle in Lesson 8. Consider this activity an extension of your exploration from the previous lesson.

Small Groups: Choose student groups based on the Wrap-Up from Lesson 8 and prompt groups to complete this activity before reviewing as a whole group.

Teacher-Led: Guide students through question 1, prompt them to answer question 2 in pairs, then review and synthesize their findings.

2. Jeffery, Katelyn, and Azia each found the area, in square feet (ft^2), in which the goat can roam. Their work is shown.

Jeffery's Work	Katelyn's Work	Azia's Work
$A = \pi r^2$ $A = \pi(20^2)$ $A = 400\pi$ ft^2	$A = \frac{1}{2}Cr$ $A = \frac{1}{2}(2\pi \cdot 20) \cdot 20$ $A = \frac{1}{2}(2 \cdot 3.14 \cdot 20)(20)$ $A \approx 1256$ ft^2	$A = \pi r^2$ $A = \pi(20^2)$ $A = \left(\frac{22}{7}\right)(400)$ $A \approx 1257.14$ ft^2

   a. Explain why Jeffery uses the equal sign $(=)$ but Katelyn and Azia both use the approximate sign $(\approx)$.
   Since 3.14 and $\frac{22}{7}$ are approximations of pi, the areas Katelyn and Azia calculated are approximations while Jeffery's shows the exact area.

   b. Harold said that he used the π button on his calculator instead of using an approximate value for π. What answer did Harold get, rounded to the hundredths place?
   $1256.64\ ft^2$

When finding the area of a circle, always write answers using the representation of π based on the directions given in the problem.

TEACHER MOVES

Use the student work to emphasize to students the importance of not rounding answers until the final step. Also have students find the value of400π using a calculator and to compare the answers using different forms of pi.

4.9.3 GUIDED PRACTICE: FINDING THE AREA OF A CIRCLE

ENRICHMENT

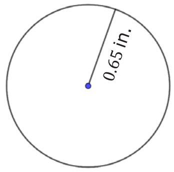
Consider using inquiry to extend student thinking as they engage independently or in small groups.

What do you notice about the steps and answers to the problem as you use the different approximations for pi?

What is different? What is similar?

4.9.3 Guided Practice: Finding the Area of a Circle

1. A circle with radius 0.65 inches (in.) is shown. Complete the table by finding the area of the circle using the given value for π.



Approximation of π	Area
3.14	$1.327 in^2$
$\frac{22}{7}$	$1.328\ in^2$
π button on the calculator	$1.327\ in^2$

TEACHER MOVES

Think-Pair-Share

Prompt each student to initially complete question 2 independently. Then, encourage students to discuss their reasoning with a partner before asking a variety of students to share their answers and justifications as a whole group.

2. Roberto wants to replace the floor on his circular skating rink but doesn't know the area. He knows the length across the skating rink through the center is 60 meters (m). He asked his two sons, Adrian and Juan, to help him determine the area. Their work, rounded to the nearest hundredth, is shown.

Adrian's Method	Juan's Method
$A = \pi \cdot 60^2$	$A = \frac{1}{2}(\pi \cdot 60) \cdot 30$
$A = \pi \cdot 3600$	$A = \pi \cdot 900$
$A = 11{,}309.73\ \text{m}^2$	$A = 2827.43\ \text{m}^2$

a. Whose method is correct?
Juan's method is correct.

b. Explain the mistake that was made to the son who made the error.
Adrian, you have to take $\frac{1}{2}$ of the diameter to get the radius before using the formula to calculate the area of the circle.

4.9.4 Your Turn!

1. Elena wants to tile the top of a circular table. The radius of the top is $14\frac{5}{8}$ inches (in.). What is the area of the table that will be tiled? Use $\frac{22}{7}$ for π.
   $672.228\ in.^2$

2. Jada painted the face of a circular clock that has a diameter of $2\frac{4}{5}$ feet (ft). What area of the clock was painted? Use 3.14 for π.
   $6.154\ ft^2$

3. A circle has a diameter that measures $7\frac{9}{10}$ centimeters (cm) in length. What is the area of the circle? Use the π on your calculator.
   $49.017\ cm^2$

4.9.4 YOUR TURN!

ENRICHMENT

Were you given the radius or the diameter? If you were given the diameter, how can you determine the radius? Why is it important to include the units with your answers? What units did you use and why?

4.9.5 Wrap-Up: Error Analysis

1. Clare and Andre are baking and frosting cookies.

Clare says:	Andre says:
"The radius of my cookie is about 3 cm, so the space for frosting is about 18.8 square centimeters (cm^2)."	"The diameter of my cookie is about 3 in., so the space for frosting is about 28.3 square inches (in^2)."

   a. Who, if anyone, correctly approximated how much space there is to spread frosting on their cookie?
   Neither correctly approximated how much space there is to spread frosting on their cookie.

   b. Write to either Clare or Andre an explanation of why their statement is incorrect.
   Clare, you used the formula to calculate the circumference of the cookie but you should have used the formula for area.

   Andre, you used the diameter instead of the radius when calculating the area.

4.9.5 WRAP-UP: ERROR ANALYSIS

TEACHER MOVES

MA.K12.MTR.4.1: Error Analysis

Students engage in analyzing the thinking and reasoning of others. For additional scaffolding, consider providing sentence starters for students to write their responses to questions 1a and 1b.

